

**DATAMITES***in association with Rubiex Internship Program***ARTIFICIAL INTELLIGENCE CAPSTONE PROJECT****PRAICP-1012 : Pneumonia Chest X-Ray Classification***A Report Submitted in Partial Fulfilment of the Requirements of the **Artificial Intelligence Capstone Project***

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Team Code	PTID-AIE-JUL-26-11142
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About the Author / Candidate Profile

This section presents a brief profile of the candidate who has authored and submitted this capstone project report, along with the administrative details under which the project was carried out as part of the DataMites Artificial Intelligence Engineer program delivered in association with the Rubiex Internship initiative.

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Project Code	PRAICP-1012
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Report Prepared On	September 2026

A Note on this Report

This document has been compiled as a comprehensive, stand-alone project report derived from the candidate's Jupyter Notebook submission for the PRAICP-1012 capstone assignment. It restates the problem statement, methodology, code-level design decisions, exploratory findings, model architectures, quantitative results and business conclusions of the original notebook in a structured, examiner-friendly format suitable for academic evaluation and portfolio presentation. Wherever personal or organisational details (such as contact information, mentor name, or batch number) are not available in the source notebook, placeholders have been used and are clearly indicated; the candidate is encouraged to update these fields before final submission.

Declaration

I, Sameeksha, hereby declare that this project report titled "PRAICP-1012: Pneumonia Chest X-Ray Classification" is a record of an original piece of work carried out by me as part of the Artificial Intelligence Capstone Project (Project Code: PRAICP-1012, Team Code: PTID-AIE-JUL-26-11142) under the DataMites Artificial Intelligence Engineer program, undertaken in association with the Rubiex Internship Project.

I further declare that the analysis, model development, results and conclusions presented in this report are based on my own work using the dataset supplied for this capstone project, and that this report has not been submitted, in whole or in part, for the award of any other certification or qualification.

All external libraries, pretrained architectures (VGG16, MobileNetV2) and reference material used in the course of this project have been duly acknowledged in the relevant sections of this report.

Signature: _____

Name: Sameeksha
Date: September 2026

Acknowledgement

I would like to express my sincere gratitude to DataMites® Innovative Solutions Pvt. Ltd. and the Rubiex Internship Project for providing me with the opportunity, mentorship, curriculum and dataset required to undertake this Artificial Intelligence Capstone Project on Pneumonia Chest X-Ray Classification.

I am grateful to my mentors and instructors for their continuous guidance on deep learning concepts, convolutional neural network design, and evaluation methodology for medical imaging problems, which greatly enriched my understanding of applying Artificial Intelligence to real-world healthcare screening challenges.

I also wish to thank my peers and the DataMites support team for their valuable feedback during the course of this project, and my family for their constant encouragement throughout the duration of this internship and capstone submission.

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Executive Summary

Pneumonia remains one of the leading causes of hospitalisation and mortality worldwide, and chest radiography (X-ray) is the most widely used, cost-effective imaging modality for its initial screening and diagnosis. However, manual interpretation of chest X-rays is time-consuming, requires specialist radiologists, and is subject to inter-observer variability, particularly in high-volume or resource-constrained healthcare settings. This capstone project, undertaken as part of the DataMites Artificial Intelligence Engineer program in association with the Rubiex Internship Project, addresses this problem by developing and comparing Convolutional Neural Network (CNN) based deep learning models capable of automatically classifying a chest X-ray image as NORMAL or PNEUMONIA.

The project uses the well-known Kermay/Mooney Chest X-Ray Pneumonia dataset, comprising 6,363 labelled images spread across training (5,218 images), validation (521 images) and test (624 images) splits, with a pronounced class imbalance in favour of the PNEUMONIA class (a ratio of approximately 2.89:1 in the training set). Three distinct CNN architectures were designed, trained and evaluated: (i) a custom CNN built and trained entirely from scratch, (ii) a transfer-learning model built on the VGG16 architecture pretrained on ImageNet, and (iii) a transfer-learning model built on the lightweight MobileNetV2 architecture. The best-performing transfer-learning model was subsequently fine-tuned by unfreezing its top convolutional layers and retraining with a substantially reduced learning rate.

All four resulting models were evaluated on a held-out test set using Accuracy, Precision, Recall, F1-Score and ROC-AUC, with particular emphasis on Recall (Sensitivity) for the PNEUMONIA class, since in a clinical screening context a false negative (missing a genuine pneumonia case) carries a far higher cost than a false positive. The custom CNN achieved the highest Recall (0.9974) but at the cost of a much lower Recall on the NORMAL class and an overall Accuracy of 0.7516, while the transfer-learning models achieved a more balanced trade-off between sensitivity and specificity, with MobileNetV2 achieving the best overall ROC-AUC after fine-tuning (0.9555). This report documents the complete workflow — from problem definition, domain analysis, data engineering and exploratory analysis, through model design, training, hyperparameter tuning, and rigorous comparative evaluation — and closes with business insights, practical deployment recommendations, project limitations and a discussion of the ethical considerations inherent in applying Artificial Intelligence to clinical decision support.

Overall, the project demonstrates that transfer learning from ImageNet-pretrained networks is a highly data-efficient and effective strategy for medical image classification tasks such as pneumonia screening, and that model selection for clinical use cases must go beyond a single metric such as Accuracy and instead be guided by the clinical cost asymmetry between false negatives and false positives.

1. Problem Statement & Project Objective

1.1 Problem Statement

Pneumonia is an infection that inflames the air sacs (alveoli) in one or both lungs, causing them to fill with fluid or purulent material. It is one of the leading infectious causes of morbidity and mortality across all age groups worldwide, and is especially dangerous for children under five and adults over sixty-five. Clinically, pneumonia is diagnosed using a combination of physical examination, patient history, blood tests and, most commonly, chest radiography (chest X-ray), in which a radiologist visually inspects the lung fields for areas of increased opacity that indicate the presence of fluid or inflammatory exudate.

Manual screening of chest X-rays, while effective, has three practical limitations that motivate this project. First, it is time-consuming: a radiologist must carefully examine each image, which is difficult to scale in high-patient-volume settings such as public hospitals or during disease outbreaks. Second, it is subject to inter-observer variability: different radiologists, or the same radiologist under fatigue or time pressure, may arrive at different conclusions for borderline cases. Third, specialist radiologists are not uniformly available, particularly in rural or resource-constrained healthcare facilities, creating a bottleneck between image acquisition and diagnosis.

The objective of this project is therefore to develop a Convolutional Neural Network (CNN) based machine learning model that can automatically classify a chest X-ray image of a patient as NORMAL or PNEUMONIA, assisting radiologists with faster and more consistent preliminary screening. Such a model is intended to function as a decision-support tool that triages incoming X-rays and flags likely-positive cases for prioritised expert review, rather than as a replacement for clinical judgement.

1.2 Project Objective

1. Build and train a CNN model that can detect Pneumonia from chest X-ray images using only the pixel data of the image, without any hand-crafted feature engineering.
2. Compare a CNN built entirely from scratch against transfer-learning based models built on top of the VGG16 and MobileNetV2 architectures, both pretrained on the ImageNet dataset.
3. Evaluate all candidate models using metrics that are appropriate for a medical screening task — namely Accuracy, Precision, Recall (Sensitivity), F1-Score and ROC-AUC — with particular attention to Recall, so that genuine Pneumonia cases (the positive class) are not missed by the model.
4. Select the best-performing model using a clinically appropriate decision rule, and provide concrete business insights, deployment recommendations and a discussion of the model's practical limitations.

1.3 Scope of the Project

The scope of this capstone project spans the complete applied machine learning lifecycle for an image classification problem: dataset acquisition and inspection, domain analysis, data cleaning, exploratory data analysis, data preprocessing and augmentation, model architecture design, model training with class-imbalance handling, hyperparameter tuning through fine-tuning, rigorous quantitative and qualitative evaluation, and finally the translation of technical results into actionable business insights. The project is implemented end-to-end in Python using TensorFlow/Keras for deep learning and

Scikit-learn for evaluation, inside a single, well-documented Jupyter Notebook, from which this report has been compiled.

2. Import of Python Libraries & Tools

Before any data can be loaded or modelled, the required Python libraries are imported. These libraries fall into five broad functional categories: general-purpose data handling, dataset acquisition, image processing, deep learning model development, and performance evaluation. Table 2.1 summarises the principal libraries used in this project and their role.

Library	Category	Purpose in this Project
os, shutil, zipfile, random, warnings	Core Python / OS	File-system navigation, dataset extraction, reproducible sampling, and suppressing non-critical warnings.
NumPy, Pandas	Data Handling	Numerical array operations and tabular summarisation of class counts, dimensions and results.
Matplotlib, Seaborn	Visualisation	Plotting class distributions, sample images, training curves, confusion matrices and ROC curves.
Pillow (PIL)	Image Processing	Opening, verifying and manipulating individual chest X-ray image files.
Requests	Dataset Acquisition	Streaming download of the compressed dataset archive from the DataMites CDN link.
TensorFlow / Keras	Deep Learning	Building, compiling, training and saving the custom CNN and transfer-learning models.
ImageDataGenerator	Data Pipeline	On-the-fly image rescaling, augmentation and batch generation directly from disk.
VGG16, MobileNetV2	Pretrained Backbones	ImageNet-pretrained convolutional bases used for transfer learning.
Scikit-learn	Evaluation	Computation of class weights, confusion matrix, classification report, and ROC/AUC metrics.

Table 2.1: Principal Python libraries used in the project and their purpose.

TensorFlow/Keras was selected as the deep learning framework for this project because it provides a mature, well-documented, high-level API for constructing both custom sequential CNNs and transfer-learning models on top of standard pretrained backbones (via `tf.keras.applications`), along with built-in utilities such as `ImageDataGenerator` for memory-efficient streaming of large image datasets, and callbacks such as `EarlyStopping`, `ReduceLROnPlateau` and `ModelCheckpoint` that are essential for training stability on a dataset of this size. Scikit-learn complements this by providing the standard classification-evaluation toolkit (`confusion_matrix`, `classification_report`, `roc_curve`, `auc`) and the `compute_class_weight` utility used to counter the class imbalance identified in Section 4.

3. Dataset Description & Domain Analysis

3.1 Dataset Acquisition

The dataset used in this project is the Chest X-Ray Images (Pneumonia vs Normal), supplied as the official DataMites Capstone Project Dataset for the PRAICP-1012 capstone assignment, and downloaded programmatically from the archive linked below.

<https://d3ilbtij3aepc.cloudfront.net/projects/CNN-PROJECT-7-11/Chest-Xray-2.zip>

The dataset contains chest X-ray images organised into two classes — NORMAL and PNEUMONIA — split into train, validation (val) and test folders. This is the standard structure of the well-known Kermay/Mooney Chest X-Ray Pneumonia dataset, on which this capstone assignment is based. To make the notebook robust to minor differences in archive layout, a directory-walking routine automatically detects whichever folder structure is present after extraction, locating any directory that directly contains both a NORMAL and a PNEUMONIA sub-folder and treating it as a valid data split.

3.2 Domain Analysis

Understanding the underlying medical imaging domain is essential before designing a CNN-based solution, since it directly informs preprocessing choices (e.g., image size, colour channels, augmentation policy) and the interpretation of results.

On a chest X-ray, radio-opaque tissue (such as bone) appears white, air appears black, and soft tissue appears in varying shades of grey. A NORMAL chest X-ray shows clear, dark (air-filled) lung fields with well-defined vascular markings and a sharply visible costophrenic angle. When pneumonia is present, the air sacs fill with fluid or inflammatory exudate instead of air, which increases the local radio-density of that region and causes it to appear as an area (or areas) of increased opacity ("whiteness" or "haziness") within the lung fields — this pattern is often referred to clinically as consolidation or an infiltrate.

Because this visual difference in texture and intensity can be subtle, localised to a small region, or partially obscured by patient positioning, rotation, or overlapping anatomical structures (such as the heart, ribs, or scapulae), pneumonia detection from chest X-rays is a good candidate problem for a Convolutional Neural Network. Unlike traditional hand-crafted image-processing pipelines that rely on manually engineered edge, texture or intensity descriptors, a CNN can learn discriminative hierarchical features — from low-level edges and textures in its early layers to high-level, disease-relevant patterns in its deeper layers — directly from raw pixel data, given a sufficiently large and representative labelled dataset.

It is also important, from a domain perspective, to recognise the clinical asymmetry of errors in this task. A False Negative (a pneumonia-positive patient incorrectly classified as NORMAL) delays treatment and can have serious health consequences, whereas a False Positive (a healthy patient incorrectly flagged as PNEUMONIA) typically results in a follow-up examination or a second opinion — an inconvenience, but a far less dangerous outcome. This asymmetry directly motivates the project's emphasis on Recall (Sensitivity) as the primary model-selection metric, discussed further in Sections 10 and 11.

4. Basic Checks on the Dataset

Before any exploratory analysis or modelling is undertaken, a set of basic sanity checks is performed to understand the size, balance and integrity of the dataset. This includes counting the number of images per class in each split, inspecting representative sample images, and scanning for corrupt or unreadable image files that could otherwise interrupt training mid-epoch.

4.1 Class Counts Across Splits

The table below reports the exact number of images per class, per split, as counted directly from the extracted dataset folders.

Split	NORMAL	PNEUMONIA	Total
Train	1,342	3,876	5,218
Validation	134	387	521
Test	234	390	624
Overall Total	1,710	4,653	6,363

Table 4.1: Image counts by class and by dataset split.

Across the entire dataset, there are 6,363 images in total. The training split alone shows a Pneumonia-to-Normal ratio of approximately 2.89 : 1, confirming a clear class imbalance weighted towards the PNEUMONIA class. Left unaddressed, this imbalance would bias a naively trained model towards over-predicting the majority (PNEUMONIA) class, inflating headline accuracy while masking poor performance on the minority NORMAL class. This observation directly motivates the class-weighting and data-augmentation strategy described in Section 5.

4.2 Sample Image Inspection

A random sample of images from each class was visually inspected as part of the basic checks. This qualitative pass confirms that the images are genuine chest radiographs correctly labelled into their respective NORMAL and PNEUMONIA folders, and gives an early, informal impression of the textural differences between the two classes that is later quantified in the Exploratory Data Analysis (Section 6).

4.3 Corrupt / Unreadable Image Check

Every image file across all three splits was opened and verified using Pillow's Image.verify() routine to detect files that are missing, truncated, or otherwise unreadable. This check identified three non-image system files (macOS ".DS_Store" metadata files that had been inadvertently included inside the class folders of the extracted archive) as the only "corrupt" entries; no genuine chest X-ray image files were found to be damaged. These three files were subsequently excluded from processing, since ImageDataGenerator only reads valid image file extensions from each class directory and therefore automatically ignores such stray system files.

5. Data Preprocessing

Raw chest X-ray images in this dataset vary considerably in native resolution and are effectively single-channel (grayscale) radiographs, even though many are saved as three-channel JPEG files. Before they can be used to train a CNN, the images must be preprocessed into a consistent, network-ready format.

5.1 Resizing and Rescaling

All images are resized to a common target size of 150×150 pixels and loaded with 3 colour channels (RGB), giving a final input shape of (150, 150, 3). Although the source X-rays are grayscale in content, loading them as 3-channel images allows the exact same data generators to feed both the custom CNN (trained from scratch) and the ImageNet-pretrained transfer-learning models, which architecturally expect 3-channel input because they were originally trained on natural RGB images. Pixel intensities, originally in the range [0, 255], are rescaled to the range [0, 1] by dividing by 255, which is standard practice for stabilising and accelerating neural network training via gradient-based optimisation.

5.2 Data Augmentation

Since the training set is imbalanced and of moderate size relative to the very deep networks used for transfer learning, data augmentation is applied to the training generator only (never to validation or test data, which must remain a faithful, unaltered representation of real-world inputs). The augmentation policy applies small, clinically plausible random transformations — including rotation (up to 10 degrees), width and height shifts (up to 10% of image dimensions), and other minor geometric perturbations — that emulate natural variation in patient positioning during X-ray acquisition without distorting the diagnostic content of the image. This exposes the network to a wider variety of spatial configurations during training, which improves generalisation and reduces overfitting to the exact pixel arrangement of the training samples.

5.3 Class Weighting to Counter Imbalance

In addition to augmentation, class weights are computed using Scikit-learn's `compute_class_weight` utility with the "balanced" heuristic, which assigns a weight to each class that is inversely proportional to its frequency in the training data. The computed weights for this dataset are shown in Table 5.1.

Class Index	Class Label	Computed Class Weight
0	NORMAL	1.9448
1	PNEUMONIA	0.6730

Table 5.1: Class weights computed to counter training-set imbalance.

Because NORMAL is the minority class in the training data, it receives a substantially higher weight (1.9448) than PNEUMONIA (0.6730). These weights are passed into the `class_weight` argument of every model's `.fit()` call, which causes the loss function to penalise misclassifications of NORMAL images more heavily than misclassifications of PNEUMONIA images, counteracting the network's natural tendency to favour the majority class purely because it appears more often during training.

5.4 Batching Strategy

Images are streamed from disk in batches of 32 using Keras' `ImageDataGenerator.flow_from_directory()` interface, rather than being loaded into memory in their entirety. This is a deliberate engineering choice: the full, uncompressed dataset occupies approximately 1.5 GB on disk, and loading every image into memory simultaneously would be both wasteful and, in constrained compute environments, infeasible. Streaming in batches keeps memory usage bounded and constant regardless of dataset size, at a small cost in I/O overhead per epoch.

6. Exploratory Data Analysis (EDA)

Exploratory Data Analysis (EDA) is performed to visually and statistically understand the dataset before committing to a specific modelling strategy. The EDA in this project covers four complementary angles: the class distribution across splits, representative sample images from each class, the distribution of original (pre-resize) image dimensions, and a comparison of average pixel intensities between the two classes.

6.1 Class Distribution Across Splits

A grouped bar chart of image counts by class (NORMAL vs. PNEUMONIA) across the Train, Validation and Test splits visually confirms the imbalance already quantified in Section 4: the PNEUMONIA class outnumbers the NORMAL class in every split, most prominently in the training set. This chart provides an intuitive, at-a-glance justification for the class-weighting strategy adopted in Section 5.3.

6.2 Sample Images per Class

A grid of sample images (four per class) from the training set was plotted for visual inspection. This confirms, qualitatively, that PNEUMONIA X-rays typically display cloudier, whiter (more opaque) regions within the lung fields, while NORMAL X-rays display clearer, darker lung fields with sharply defined vascular markings — consistent with the domain analysis in Section 3.2.

6.3 Distribution of Original Image Dimensions

Since chest X-rays in this dataset were sourced from multiple clinical acquisition systems, their native (pre-resize) dimensions vary considerably. A sample of 300 images was drawn and measured, yielding the summary statistics in Table 6.1.

Dimension	Minimum (px)	Maximum (px)	Mean (px)
Width	437	2,619	1,353
Height	144	2,628	1,001

Table 6.1: Summary statistics of original image dimensions (sampled, $n = 300$).

This wide spread in native resolution — spanning roughly a six-fold difference between the smallest and largest images in both width and height — confirms the necessity of the uniform resizing step performed in Section 5.1, since a CNN's fully-connected and convolutional layers require fixed-size input tensors.

6.4 Average-Image Intensity Comparison

To obtain a more quantitative view of the textural difference between the two classes, an "average image" was constructed for each class by resizing 200 sampled images per class to a common size, converting them to grayscale, and averaging their pixel intensities element-wise. The resulting average NORMAL and average PNEUMONIA images show a visibly different overall intensity/opacity pattern in the central lung-field region, providing quantitative support for the hypothesis that a CNN can learn to discriminate between the two classes based on textural and intensity cues alone.

6.5 EDA Summary (Task 1 Conclusion)

1. The dataset is organised into NORMAL and PNEUMONIA classes across train/validation/test splits, with the training set showing a noticeable class imbalance in favour of PNEUMONIA (approximately 2.89 Pneumonia images for every 1 Normal image) — this was addressed later using class weights and data augmentation.
2. Sample images confirm that PNEUMONIA X-rays typically show cloudier, whiter (opaque) regions within the lung fields compared to the clearer, darker lung fields seen in NORMAL X-rays.
3. Image dimensions vary considerably across the dataset (width from 437 to 2,619 pixels; height from 144 to 2,628 pixels), which confirms the need to resize every image to a fixed size (150 × 150) before feeding them into the CNN.
4. The average-image comparison shows a visibly different overall intensity/opacity pattern between the two classes, supporting the choice of a CNN-based approach that can learn these texture differences automatically.

7. Data Preparation for Modelling

7.1 Generators & Class Weights Recap

Unlike a tabular dataset, where a train/validation/test split must typically be created manually (e.g., using `train_test_split`), the split for this dataset is already provided by its own folder structure (train / val / test subfolders per class). The `ImageDataGenerator` objects configured in Section 5 stream batches of resized, rescaled (and, for the training generator only, augmented) images directly from disk, which avoids having to load the full ~1.5 GB of image data into memory at once. The `class_weights` computed in Section 5.3 are passed into every model's `.fit()` call to counter the class imbalance identified during the basic checks.

One practical data-engineering issue surfaced during preparation: the publicly distributed version of this dataset ships with an extremely small original validation folder (containing only a handful of images per class), which is not statistically sufficient to reliably monitor training progress or to make early-stopping decisions. The notebook automatically detects this condition and, where the original validation folder is too small to be useful, carves out a proper validation split (10% of the training data) directly from the training folder instead, ensuring that validation metrics reported during training are statistically meaningful.

7.2 Generator Configuration Summary

Parameter	Value
Target Image Size	150 × 150 pixels
Input Shape	(150, 150, 3)
Colour Mode	RGB (3-channel)
Batch Size	32
Training Images (after split adjustment)	5,216
Validation Images (after split adjustment)	520
Test Images	624
Steps per Epoch (Train)	163
Validation Steps	16
Test Steps	20
Class Index Mapping	NORMAL = 0, PNEUMONIA = 1

Table 7.1: Final generator configuration used for all three model architectures.

A shared `evaluate_model()` helper function was defined at this stage and reused consistently for every trained model. It runs the model's predictions over the entire test generator, applies a 0.5 probability threshold to convert predicted probabilities into binary class predictions, and computes Accuracy, Precision, Recall, F1-Score and ROC-AUC in a single, consistent pass — ensuring that every model in this report (the custom CNN, VGG16, MobileNetV2, and the fine-tuned model) is compared on an identical, like-for-like basis.

8. Model Building & Training

To classify the chest X-rays, three distinct Convolutional Neural Network architectures were designed, trained and compared. Each model uses EarlyStopping (to halt training once validation loss stops improving, restoring the best-performing weights) and ReduceLROnPlateau (to automatically lower the learning rate when training progress stalls), and each is checkpointed to disk via ModelCheckpoint so that the best-performing weights are always retained.

1. Model 1 — Custom CNN (from scratch): a straightforward convolutional network trained end-to-end using only this dataset, with no pretrained weights.
2. Model 2 — Transfer Learning with VGG16: the convolutional base of VGG16 (pretrained on the 1.4-million-image ImageNet dataset) is frozen and used as a fixed feature extractor, with a new, lightweight classification head trained on top.
3. Model 3 — Transfer Learning with MobileNetV2: a lightweight, computationally efficient pretrained network, again used as a frozen feature extractor with a custom classification head, chosen to explore the accuracy/efficiency trade-off relative to the much larger VGG16.

8.1 Model 1 — Custom CNN (from Scratch)

The custom CNN is a Sequential Keras model consisting of four convolutional blocks, each composed of a Conv2D layer (with 32, 64, 128 and 128 filters respectively, using 3×3 kernels and ReLU activation), followed by Batch Normalization (to stabilise and accelerate training by normalising layer activations) and MaxPooling2D (to progressively down-sample the spatial resolution and build translation-invariant features). The convolutional feature maps are then flattened and passed through fully connected (Dense) layers with Dropout regularisation, culminating in a single sigmoid output neuron that produces the probability of the PNEUMONIA class.

Architecture Property	Value
Convolutional Blocks	4 (32 → 64 → 128 → 128 filters)
Regularisation	Batch Normalization + Dropout
Output Layer	Dense(1, activation='sigmoid')
Total Parameters	1,848,385
Trainable Parameters	1,847,681
Non-Trainable Parameters	704
Optimizer	Adam
Loss Function	Binary Cross-Entropy
Epochs (max, with Early Stopping)	15

Table 8.1: Architecture and training configuration of the custom CNN (Model 1).

Being trained entirely from scratch, this model must learn all of its convolutional filters purely from the ~5,200 training images available, without the benefit of any prior visual knowledge. Its training used EarlyStopping (patience = 4 epochs on validation loss), ReduceLROnPlateau (halving the learning rate after 2 stagnant epochs, down to a floor of 1e-6), and ModelCheckpoint to persist the best weights observed.

8.2 Model 2 — Transfer Learning: VGG16

VGG16 is a deep, 16-layer convolutional network originally developed by the Visual Geometry Group at Oxford and pretrained on the ImageNet dataset (over 1.4 million natural images across 1,000 classes). In this project, the VGG16 convolutional base is loaded with its ImageNet weights and its trainable attribute is set to False, freezing all of its convolutional filters so that only a new classification head is trained. This head consists of a GlobalAveragePooling2D layer (which condenses each feature map into a single value, drastically reducing the parameter count relative to a Flatten layer), a Dense(128, activation='relu') layer, a Dropout(0.4) layer for regularisation, and a final Dense(1, activation='sigmoid') output neuron.

Architecture Property	Value
Backbone	VGG16 (ImageNet-pretrained, frozen)
Classification Head	GlobalAveragePooling2D → Dense(128, relu) → Dropout(0.4) → Dense(1, sigmoid)
Total Parameters	14,780,481
Trainable Parameters	65,793
Non-Trainable (Frozen) Parameters	14,714,688
Optimizer	Adam
Loss Function	Binary Cross-Entropy
Epochs (max, with Early Stopping)	15

Table 8.2: Architecture and training configuration of the VGG16 transfer-learning model (Model 2).

Because more than 99.5% of VGG16's parameters remain frozen, only 65,793 parameters are actually updated during training — a small fraction of the 1.8 million trained from scratch in Model 1. This dramatically reduces the risk of overfitting on a comparatively small, domain-specific medical imaging dataset, since the network leverages visual features (edges, textures, shapes) already learned from ImageNet rather than re-deriving them from limited chest X-ray data.

8.3 Model 3 — Transfer Learning: MobileNetV2

MobileNetV2 is a lightweight convolutional architecture designed for mobile and edge deployment, built around depth-wise separable convolutions and inverted residual blocks that dramatically reduce the number of parameters and floating-point operations relative to classic architectures such as VGG16, while retaining strong ImageNet classification accuracy. It is used here in exactly the same transfer-learning configuration as VGG16: the pretrained convolutional base is frozen, and the identical classification head (GlobalAveragePooling2D → Dense(128, relu) → Dropout(0.4) → Dense(1, sigmoid)) is trained on top, allowing a fair, architecture-controlled comparison between the two transfer-learning backbones.

Architecture Property	Value
Backbone	MobileNetV2 (ImageNet-pretrained, frozen)
Classification Head	GlobalAveragePooling2D → Dense(128, relu) → Dropout(0.4) → Dense(1, sigmoid)
Total Parameters	2,422,081

Architecture Property	Value
Trainable Parameters	164,097
Non-Trainable (Frozen) Parameters	2,257,984
Optimizer	Adam
Loss Function	Binary Cross-Entropy
Epochs (max, with Early Stopping)	15

Table 8.3: Architecture and training configuration of the MobileNetV2 transfer-learning model (Model 3).

MobileNetV2's total parameter count (2.42 million) is roughly six times smaller than VGG16's (14.78 million), while its per-step training time observed during this project (approximately 600–850 milliseconds per step) was also substantially faster than VGG16's (approximately 6–7 seconds per step) on the same hardware, illustrating the well-documented efficiency advantage of MobileNetV2-style architectures for deployment-constrained environments.

9. Hyperparameter Tuning (Fine-Tuning the Best Transfer-Learning Model)

Rather than a generic grid or random search over hyperparameters — which is prohibitively expensive for CNNs of this size given the training time per epoch — the standard and most effective hyperparameter-tuning strategy for transfer learning is fine-tuning: the best-performing frozen transfer-learning model is identified, a controlled number of its top convolutional layers are unfrozen, and the model is retrained for a small number of additional epochs using a much smaller learning rate. This allows the pretrained filters to adapt slightly to the specific texture and intensity patterns found in chest X-rays, without catastrophically overwriting the general-purpose visual features learned from ImageNet.

9.1 Selecting the Base Model for Fine-Tuning

Before fine-tuning, the two frozen transfer-learning models (VGG16 and MobileNetV2) were compared on validation/test ROC-AUC. MobileNetV2 (Transfer Learning) was identified as the stronger of the two frozen transfer-learning models and was therefore selected as the candidate for fine-tuning.

9.2 Fine-Tuning Configuration

The top 30 layers of the MobileNetV2 base were unfrozen (`fine_tune_base.trainable = True`, with all layers up to `len(base.layers) - 30` kept frozen), while the earlier, more generic convolutional layers remained frozen to preserve low-level ImageNet features such as edges and colour gradients. The model was then re-compiled with the Adam optimizer at a substantially reduced learning rate of 1×10^{-5} (compared to the default 1×10^{-4} used for the initial frozen-base training), since fine-tuning pretrained weights with a large learning rate risks destroying the useful representations already learned. Training proceeded for up to 8 additional epochs, again governed by `EarlyStopping` and `ReduceLROnPlateau`.

Fine-Tuning Parameter	Value
Base Model Fine-Tuned	MobileNetV2 (Transfer Learning)
Layers Unfrozen	Top 30 layers of the MobileNetV2 base
Fine-Tuning Learning Rate	1×10^{-5}
Original (Frozen-Base) Learning Rate	1×10^{-4}
Additional Epochs (max)	8
Final Training Accuracy Observed	$\approx 95.17\%$
Final Validation Accuracy Observed	$\approx 94.92\%$

Table 9.1: Fine-tuning configuration applied to the best frozen transfer-learning model.

This fine-tuning strategy is, functionally, a targeted and computationally efficient alternative to a full hyperparameter grid search: rather than exhaustively searching over learning rates, batch sizes, and architectural depth from a blank slate, the search space is deliberately narrowed to the single highest-leverage decision for transfer learning — how much of the pretrained network to re-open for adaptation, and at what learning rate — which is both standard practice in the field and appropriate given the compute budget of this capstone.

10. Model Comparison Report

After training and evaluating all four models (the custom CNN, VGG16, MobileNetV2, and fine-tuned MobileNetV2) using the shared `evaluate_model()` helper on the identical, untouched test set of 624 images, their performance is compared using Accuracy, Precision, Recall, F1-Score and ROC-AUC. For a medical screening task such as this, Recall (Sensitivity) on the PNEUMONIA class is arguably the most clinically important metric, since a False Negative (telling a sick patient they are healthy) is far more costly than a False Positive (telling a healthy patient they may need a follow-up). Accordingly, the comparison table is deliberately sorted by Recall rather than by raw Accuracy.

Model	Accuracy	Precision	Recall	F1-Score	ROC-AUC
Custom CNN	0.7516	0.7164	0.9974	0.8339	0.9456
MobileNetV2 (Fine-Tuned)	0.8734	0.8575	0.9564	0.9042	0.9555
MobileNetV2 (Transfer Learning)	0.8846	0.8841	0.9385	0.9104	0.9510
VGG16 (Transfer Learning)	0.8798	0.9156	0.8897	0.9025	0.9363

Table 10.1: Test-set performance comparison across all four candidate models, sorted by Recall (descending).

10.1 Interpreting the Comparison

The custom CNN achieves the single highest Recall of any model (0.9974), meaning it correctly identifies 99.74% of all genuine PNEUMONIA cases in the test set — an extremely low False Negative rate. However, this comes at the cost of the lowest Precision (0.7164) and lowest overall Accuracy (0.7516) among the four models, indicating that it achieves this near-perfect sensitivity partly by aggressively over-predicting the PNEUMONIA class, including on a substantial number of genuinely NORMAL images (explored quantitatively in Section 11).

By contrast, the three transfer-learning models (VGG16, MobileNetV2, and fine-tuned MobileNetV2) offer a more balanced trade-off: each achieves an overall Accuracy above 87%, Precision above 0.85, and Recall above 0.88, together with the three highest ROC-AUC scores in the comparison (0.9363–0.9555). Among these, fine-tuned MobileNetV2 achieves the best overall ROC-AUC (0.9555) and a strong Recall (0.9564), while frozen MobileNetV2 achieves the best F1-Score (0.9104) and best raw Accuracy (0.8846). VGG16 achieves the highest Precision of any model (0.9156) but the lowest Recall among the transfer-learning models (0.8897).

This comparison illustrates a recurring theme in applied machine learning: no single model dominates on every metric simultaneously, and the choice of a "best" model is inseparable from the business or clinical objective the model is meant to serve — a theme explored further in Sections 11 and 12.

10.2 Training Behaviour Across Models

Validation accuracy and validation loss curves were plotted across training epochs for all four models. The custom CNN's validation loss showed comparatively higher volatility across early epochs (consistent with training a network from scratch on a moderately sized dataset), while both transfer-learning models converged more smoothly and reached a given validation accuracy in fewer epochs,

reflecting the head start provided by ImageNet-pretrained weights. MobileNetV2 additionally converged noticeably faster in wall-clock time per epoch than VGG16, consistent with its smaller parameter count and lighter-weight depth-wise separable convolutions.

11. Best Model — Detailed Evaluation

Following the Recall-first selection rule adopted for this clinical screening use case (Section 10), the custom CNN is identified as the “best performing model” by the model comparison table and is subjected to a deeper, more granular evaluation using a confusion matrix, a full per-class classification report, and an ROC curve. This deeper analysis is essential, because a single top-line Recall figure can obscure important trade-offs that only become visible at the level of individual classes — as the results below demonstrate.

11.1 Confusion Matrix

The confusion matrix for the custom CNN on the 624-image test set (234 NORMAL, 390 PNEUMONIA) shows that the model correctly identifies almost every genuine PNEUMONIA case, at the cost of misclassifying a substantial proportion of genuinely NORMAL X-rays as PNEUMONIA. This pattern is confirmed quantitatively by the classification report below.

11.2 Classification Report

Class	Precision	Recall	F1-Score	Support
NORMAL	0.99	0.34	0.51	234
PNEUMONIA	0.72	1.00	0.83	390
Accuracy (overall)			0.75	624
Macro Average	0.85	0.67	0.67	624
Weighted Average	0.82	0.75	0.71	624

Table 11.1: Per-class classification report for the custom CNN (best model by Recall) on the test set.

This table reveals a critical nuance that is invisible in the single aggregate Recall figure reported in Section 10: while the custom CNN achieves a Recall of 1.00 (100%) on the PNEUMONIA class — it misses zero genuine pneumonia cases in the test set — its Recall on the NORMAL class is only 0.34. In other words, of the 234 genuinely healthy patients in the test set, the model correctly identifies only about 34% of them as NORMAL, and incorrectly flags the remaining $\approx 66\%$ as PNEUMONIA. Its Precision on the NORMAL class is very high (0.99), meaning that on the rare occasions it does predict NORMAL, it is almost always correct — but it predicts NORMAL far too rarely.

Clinically, this describes a model with extremely high sensitivity (it will not miss a sick patient) but comparatively low specificity (it will incorrectly alarm a large fraction of healthy patients). Whether this trade-off is acceptable depends entirely on the intended deployment context, and is discussed further in Section 12 (Business Insights).

11.3 ROC Curve

The Receiver Operating Characteristic (ROC) curve for the custom CNN plots the True Positive Rate against the False Positive Rate across all possible classification thresholds, yielding an Area Under the Curve (ROC-AUC) of 0.9456. This is a strong discriminative score, and importantly it is a threshold-independent measure of the model's underlying ability to separate the two classes by predicted probability — unlike the 0.5-threshold classification report above, which reflects only one specific operating point on this curve. This distinction is significant: the poor NORMAL-class Recall observed at the default 0.5 threshold does not necessarily reflect a poor underlying model; it may

instead reflect a sub-optimally chosen decision threshold, which could potentially be raised (e.g. above 0.5) to trade some Recall for improved Precision and NORMAL-class specificity, a possibility discussed further in Sections 12 and 17.

11.4 Qualitative Review of Individual Predictions

A random sample of individual test-set predictions, drawn from both correctly and incorrectly classified images, was plotted alongside their true and predicted labels for a final qualitative sanity check. This step is a useful complement to aggregate metrics, as it allows a human reviewer to visually confirm that the model's errors are concentrated on genuinely ambiguous or borderline images, rather than on clearly obvious cases, which would indicate a more fundamental problem with the model or data pipeline.

12. Business Insights

Beyond the purely technical evaluation of Sections 10 and 11, this project yields several actionable business and clinical-deployment insights, summarised below.

12.1 Screening Assistance, Not Replacement

The trained models achieve strong Recall on the PNEUMONIA class, meaning they can act as a fast, automated first-pass screening tool that flags likely-positive chest X-rays for a radiologist to prioritise and confirm. Critically, such a model should support, not replace, clinical diagnosis: it is best positioned as a triage layer that helps a radiology department allocate its limited expert attention to the cases most likely to require it, rather than as an autonomous diagnostic authority.

12.2 Transfer Learning is More Data-Efficient

The transfer-learning models (VGG16 and MobileNetV2), which start from ImageNet-pretrained filters, generally reached strong, stable performance with a more favourable balance of Precision and Recall than the custom CNN trained purely from scratch, and did so while updating orders of magnitude fewer trainable parameters (65,793 for VGG16 and 164,097 for MobileNetV2, versus 1,847,681 for the custom CNN). This demonstrates that pretrained visual features, learned from millions of natural images, transfer well to the medical-imaging domain even though chest X-rays are visually very different from ImageNet's everyday photographs — a finding consistent with a substantial body of published medical-imaging literature.

12.3 Recall Over Accuracy for Triage, but Context Matters

Because missing a true Pneumonia case (a False Negative) is clinically more dangerous than a false alarm, the model comparison in this project is deliberately sorted by Recall rather than raw Accuracy. However, Section 11 shows that pursuing Recall to its absolute maximum (as the custom CNN does) can come at a steep cost to specificity, generating a large number of false alarms on healthy patients. In a real deployment, the classification threshold should be tuned — and can be lowered or raised from the default 0.5 — to trade some Precision for higher Recall, or vice versa, according to the specific hospital's risk tolerance, radiologist capacity, and the downstream cost of a false alarm (e.g. an unnecessary follow-up scan, patient anxiety, or radiologist time spent re-reviewing a false alert).

12.4 Deployment Consideration: Model Size vs. Accuracy

MobileNetV2 is substantially smaller (2.42 million parameters) and faster to run than VGG16 (14.78 million parameters), while matching or exceeding VGG16 on every evaluation metric in this project. This makes MobileNetV2 a strong candidate for deployment on lower-cost hardware, in resource-constrained clinical settings, or at the network edge (for example, embedded directly into an X-ray machine's on-device software in a rural clinic with limited connectivity to a central server), provided its sensitivity is validated as clinically sufficient for the target population.

12.5 Recommended Operating Model

1. Deploy a transfer-learning model (MobileNetV2, fine-tuned) rather than the from-scratch CNN as the primary screening model, given its more balanced Precision/Recall profile and its markedly better handling of the NORMAL class.

2. Present the model's output as a probability score together with a recommended threshold, rather than a hard binary label, so that radiologists retain visibility into the model's confidence and can apply their own clinical judgement on borderline cases.
3. Periodically re-calibrate the decision threshold against locally observed disease prevalence and radiologist feedback, since a threshold tuned on this specific dataset may not transfer perfectly to a different hospital's patient population or X-ray equipment.
4. Treat the model as a continuously monitored clinical decision-support tool, with a feedback loop that captures radiologist overrides and disagreements to inform periodic retraining.

13. Challenges Faced

13.1 Large Dataset Size

At approximately 1.5 GB, the raw dataset is too large to attach directly to the notebook or to load entirely into memory. This was addressed by using Keras' `flow_from_directory` generators, which stream images from disk in fixed-size batches during training rather than requiring the full dataset to reside in memory simultaneously.

13.2 Class Imbalance

The training set contains noticeably more PNEUMONIA images than NORMAL images (a ratio of approximately 2.89:1). This was addressed with class weighting (via Scikit-learn's `compute_class_weight`) so that the loss function does not simply learn to predict the majority class to minimise error, combined with data augmentation applied to the training generator to further improve generalisation on the minority class.

13.3 An Unusably Small Original Validation Set

The commonly distributed version of this dataset ships with only a handful of validation images — far too few to reliably monitor training progress, tune early-stopping decisions, or trust reported validation metrics. The notebook automatically detects this condition and, in that case, carves out a statistically adequate validation split (10% of the training data) from the training folder itself, ensuring that validation-based decisions during training are made on a meaningful sample size.

13.4 Risk of Overfitting on a Relatively Small, Domain-Specific Dataset

Medical imaging datasets, including this one, are dramatically smaller than the datasets (such as ImageNet, with over a million images) used to pretrain networks like VGG16 and MobileNetV2. This creates a real risk of overfitting, particularly for the custom CNN trained entirely from scratch. To mitigate this risk, the project combines several complementary regularisation strategies: aggressive but clinically plausible data augmentation, Dropout layers in the classification heads, Batch Normalization in the custom CNN, and EarlyStopping to halt training as soon as validation performance stops improving.

13.5 Long Per-Epoch Training Times for the VGG16 Backbone

Training the VGG16-based model was observed to take considerably longer per epoch (on the order of 1,000–1,200 seconds per epoch on the available hardware, which had no GPU acceleration) than either the custom CNN or MobileNetV2, due to VGG16's much larger convolutional base performing a forward pass over every training batch even while frozen. This was accepted as a necessary cost for including VGG16 as a comparison point, but it reinforces the practical deployment argument in favour of the more lightweight MobileNetV2 architecture (Section 12.4).

14. Final Conclusion

14.1 Project Objective

The objective of this project was to build a machine learning model capable of classifying chest X-ray images as NORMAL or PNEUMONIA, to support faster and more consistent clinical screening in a way that could realistically be deployed as a decision-support tool.

14.2 Exploratory Data Analysis

EDA confirmed a visible textural difference between NORMAL and PNEUMONIA X-rays, considerable variability in raw image dimensions (requiring a uniform resizing step), and a class imbalance in the training data (approximately 2.89:1 in favour of PNEUMONIA) that needed to be corrected for during modelling.

14.3 Modelling

A custom CNN and two transfer-learning models (VGG16 and MobileNetV2) were trained and compared on an identical, held-out test set. The best-performing frozen transfer-learning model, MobileNetV2, was further fine-tuned by unfreezing its top 30 layers and retraining with a substantially reduced learning rate (1×10^{-5}), which improved its ROC-AUC further to 0.9555, the best of any model evaluated in this project.

14.4 Outcome

Ranked strictly by Recall on the PNEUMONIA class, the custom CNN achieves the top position (0.9974), making it, in principle, the most sensitive screening candidate; however, its very poor Recall on the NORMAL class (0.34) means it generates a high rate of false alarms and would not be an appropriate choice for unsupervised deployment. The three transfer-learning models offer a more clinically balanced and practically deployable alternative, with fine-tuned MobileNetV2 achieving the strongest overall discriminative power (ROC-AUC 0.9555) alongside a still-strong Recall (0.9564). Taken together, the project's final recommendation is that a fine-tuned, transfer-learning-based MobileNetV2 model is the most suitable candidate for an assistive screening tool, subject to further clinical validation, threshold calibration against local disease prevalence, and ongoing performance monitoring after deployment.

15. Project Links

The following links provide access to the complete project resources, including the source code, dataset, and supporting documentation.

15.1 GitHub Repository

Access to the complete source code, Jupyter Notebook, README, and project files:

https://github.com/Sameekshagithub/PRAICP-1012-Pneumonia_chest_x-ray_classification.git

15.2 Google Drive

Access to the accompanying presentation (PPT) and this report:

https://drive.google.com/drive/folders/1YP1raqGiWEfrQ1cQYbN00vaBCktRPx9m?usp=drive_link

16. Ethical Considerations & Limitations

16.1 Ethical Considerations

1. **Clinical Accountability:** An AI screening model must never be the final decision-maker in a diagnostic pathway. All predictions in this project are intended to route cases to a qualified radiologist for confirmation, preserving human accountability for every clinical decision.
2. **Dataset Representativeness:** The dataset used here originates from a specific patient population and set of imaging equipment (the paediatric cohort underlying the original Kermany/Mooney dataset). A model trained on this data may not generalise equally well to adult populations, different demographics, or X-ray machines with different calibration, and should be re-validated before deployment in a different clinical setting.
3. **False Alarm Burden:** As shown in Section 11, aggressively optimising for Recall can impose a significant false-alarm burden on healthy patients and on radiologist workload. Ethical deployment requires an explicit, clinically negotiated trade-off between sensitivity and specificity, rather than a default optimisation for a single metric.
4. **Data Privacy:** Chest X-ray images are sensitive protected health information. Any real-world deployment of a model such as this must comply with applicable data protection and health-information regulations (such as HIPAA in the United States or India's Digital Personal Data Protection Act), including secure storage, access controls, and patient consent where required.

16.2 Limitations of this Project

1. **Single-Dataset Evaluation:** All models were trained and evaluated on a single public dataset. Performance on external, out-of-distribution X-rays (from different hospitals, scanners, or patient populations) has not been tested and cannot be assumed to match the results reported here.
2. **Binary Classification Only:** The project distinguishes only between NORMAL and PNEUMONIA, and does not differentiate between bacterial and viral pneumonia, nor does it detect other thoracic pathologies (such as tuberculosis, pleural effusion, or lung nodules) that may co-occur or be confused with pneumonia on a chest X-ray.
3. **Default Decision Threshold:** All classification-report metrics in Section 11 are reported at the default 0.5 probability threshold; the discussion in Sections 11.3 and 12.3 shows that a different threshold could materially change the observed Precision/Recall trade-off, and this was not exhaustively explored as part of the current scope.
4. **No External Clinical Validation:** This project constitutes a technical proof-of-concept developed as part of an academic capstone. It has not undergone prospective clinical trial validation, regulatory review, or approval, and must not be interpreted as a certified diagnostic device.

17. Future Scope & Recommendations

1. **Threshold Optimisation:** Systematically sweep the classification threshold and plot Precision-Recall trade-off curves to select an operating point that is explicitly calibrated to a target clinical sensitivity/specificity, rather than relying on the default 0.5 cut-off.
2. **Explainability:** Incorporate visual explanation techniques such as Grad-CAM (Gradient-weighted Class Activation Mapping) to overlay heatmaps on chest X-rays showing which image regions most influenced the model's prediction, increasing radiologist trust and enabling error auditing.
3. **Multi-Class Extension:** Extend the current binary NORMAL/PNEUMONIA classifier into a multi-class or multi-label model that can additionally distinguish bacterial versus viral pneumonia, and flag other common thoracic findings.
4. **Cross-Institutional Validation:** Validate the model's performance on chest X-rays sourced from different hospitals, scanners and patient demographics to test and improve generalisation before any real-world pilot deployment.
5. **Ensembling:** Explore ensembling the custom CNN with the transfer-learning models (for example, by averaging predicted probabilities) to combine the custom CNN's high sensitivity with the transfer-learning models' better specificity.
6. **Model Compression for Edge Deployment:** Apply post-training quantisation or pruning to the fine-tuned MobileNetV2 model to further reduce its footprint for deployment on low-power edge devices in rural or resource-constrained clinics.
7. **Prospective Clinical Study:** Design a prospective study in collaboration with a radiology department to measure the model's real-world impact on screening turnaround time and diagnostic concordance with expert radiologists.

18. References

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Appendix A: Glossary of Terms

CNN (Convolutional Neural Network): A class of deep neural networks that use convolutional layers to automatically learn spatial hierarchies of features from image data.

Transfer Learning: A technique in which a model developed for one task (e.g., ImageNet classification) is reused, in whole or in part, as the starting point for a model on a second, related task (e.g., pneumonia detection).

ImageNet: A large-scale visual database of over 1.4 million labelled natural images across 1,000 categories, widely used to pretrain computer-vision backbones.

Data Augmentation: The practice of applying random, label-preserving transformations (rotation, shifting, flipping, etc.) to training images to artificially increase data diversity and reduce overfitting.

Class Weighting: A technique that assigns higher loss penalties to under-represented classes during training, to counter the effects of class imbalance.

Accuracy: The proportion of all predictions (both classes) that are correct.

Precision: Of all instances predicted as the positive class (PNEUMONIA), the proportion that are truly positive.

Recall (Sensitivity): Of all instances that are truly positive (PNEUMONIA), the proportion that are correctly identified by the model.

Specificity: Of all instances that are truly negative (NORMAL), the proportion that are correctly identified by the model.

F1-Score: The harmonic mean of Precision and Recall, providing a single balanced metric when both false positives and false negatives matter.

ROC-AUC: The Area Under the Receiver Operating Characteristic Curve; a threshold-independent measure of a classifier's ability to separate two classes.

Fine-Tuning: The process of unfreezing some or all layers of a pretrained model and continuing training (typically at a low learning rate) to adapt it further to a new dataset.

Early Stopping: A regularisation technique that halts training once a monitored metric (e.g., validation loss) stops improving, to prevent overfitting.

Batch Normalization: A layer that normalises its inputs across a mini-batch, which stabilises and accelerates neural network training.

Dropout: A regularisation technique that randomly deactivates a fraction of neurons during training to reduce overfitting and co-dependency between neurons.

Appendix B: Environment & Reproducibility Notes

Component	Detail
Programming Language	Python 3
Deep Learning Framework	TensorFlow 2.20.0 (Keras high-level API)
Hardware Used	CPU only (no GPU acceleration available in the training environment)
Notebook Environment	Jupyter Notebook
Dataset Size (uncompressed)	≈ 1.5 GB
Random Seed / Sampling	Fixed sampling used for EDA visualisations to aid reproducibility
Model Checkpointing	Best weights per model saved via ModelCheckpoint (.h5 format)

The absence of GPU acceleration materially affected per-epoch training time, particularly for the VGG16-based model (observed at approximately 6–7 seconds per training step). Readers reproducing this project on GPU-enabled hardware should expect substantially shorter training times, though final model metrics should remain broadly consistent given the same data, architecture, and random augmentation policy, subject to normal run-to-run variance inherent in stochastic neural network training.

For full reproducibility, refer to the original Jupyter Notebook (PRAICP_1012_Pneumonia_chest_x_ray_classification.ipynb) linked in Section 15, which contains the complete, executable source code corresponding to every section of this report.